Electronics Engineer

Embedded Systems & Microcontroller Specialist

ATmega328P Arduino Proteus C/C++ Firmware

Pakistan | **Upwork Freelancer**

4 Months	1	4-5h
Embedded Exp.	MCU Project Completed	Avg. Response Time

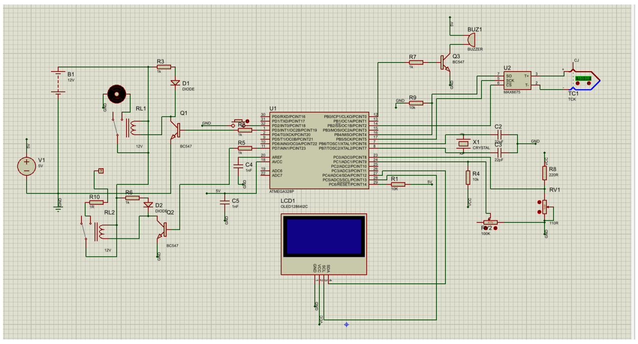
Professional Summary

I am an embedded systems engineer focused on designing, simulating, and deploying microcontroller-based hardware and firmware. I strictly follow a **Design** → **Simulate (Proteus)** → **Flash** → **Validate** pipeline. My expertise lies in **ATmega328P / Arduino** platforms, bare-metal C++ firmware, ADC measurement loops, and safety-critical Finite State Machines (FSM).

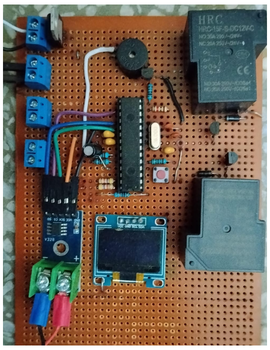
ATmega328P | Arduino IDE | Proteus Simulation | Bare-metal C/C++ | ADC & Sensors | FSM Firmware | SPI / I2C | PCB Hardware

Featured Project — AJM Industrial Boiler Control System

The **AJM Boiler Control System** is a safety-critical industrial firmware project built around the **ATmega328P**. The system autonomously manages a high-voltage water pump and heater element using a deterministic Finite State Machine (FSM), continuously monitoring water level, pressure, and temperature.



Proteus Software-in-the-Loop Simulation



Physical ATmega328P Hardware Build

Technical Implementation Highlights:

- **Strict Linear FSM:** Engineered a non-blocking state machine handling 9 distinct operational phases without relying on asynchronous delays, preventing relay clashing.
- **Debounced Safety Halts:** Written custom logic to detect physical wire-cuts (e.g., MAX6675 SPI disconnects, ADC drops) with a 2-second debounce window to filter out electrical EMI noise.
- **Thermal Hysteresis:** Heater safely locks out at $>170°C$ or >70 PSI, but smartly waits for the system to drop to $\leq 160°C$ and ≤ 60 PSI before resuming, preventing relay chatter.
- **Memory Optimised UI:** Built an industrial SSD1306 OLED interface featuring dynamic arc gauges and procedural boot animations, successfully optimising C++ functions (replacing `snprintf` with `itoa`) to keep firmware under the strict 32KB flash limit.

Portfolio prepared for Upwork proposal submission. All projects are original work. 10/04/2026